

AI-Powered Internship Application Tracker on AWS

Project Idea for Application Development on AWS

Project Summary: This project develops a cloud-based web application that helps students manage internship applications, upload documents, track application progress, and use AI to compare their CV with job descriptions. The system is deployed on AWS and integrates services such as S3, RDS, API Gateway, EC2 or Lambda, and CloudWatch.

1 Project Title

AI-Powered Internship Application Tracker on AWS

2 Core System Features

The system provides basic web application functions for managing internship applications.

- User login and registration
- Add internship application
- Track application status
- Upload CV, transcript, and certificate
- Store uploaded documents in Amazon S3
- Store application data in Amazon RDS
- View dashboard and application statistics

3 AI Features

The AI module helps users improve their internship preparation and evaluate how well their CV matches a job description.

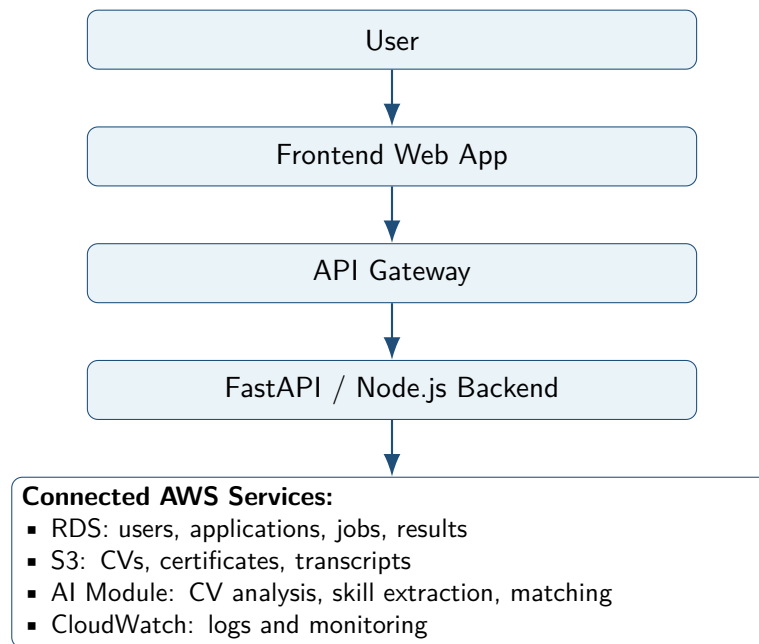
- Extract skills from CV
- Compare CV with job description
- Generate a matching score
- Show missing skills

- Generate cover letter
- Generate interview questions

This project is strong because it is still a complete web application, but it also includes AI features that make it more useful, modern, and impressive for demonstration.

4 Simple AWS Architecture

The system can be designed with the following AWS-based architecture.



5 AI Implementation Approaches

5.1 Easy Version: LLM API

The easiest approach is to use an LLM API to analyze the CV and job description.

Input:

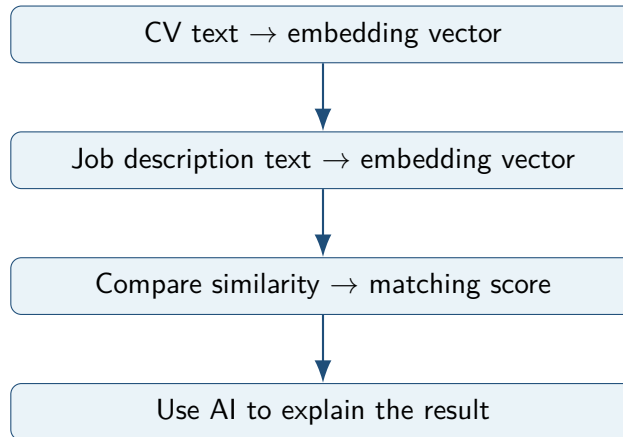
- CV text
- Job description text

Output:

- Matching score
- Matched skills
- Missing skills
- Suggested improvements

5.2 More Technical Version: Embeddings and Cosine Similarity

A more technical approach is to use embeddings and cosine similarity to compare the CV and job description.



6 Suggested Database Tables

The system can use the following database tables.

Table	Purpose
users	Stores user information such as name, email, password hash, and account role.
applications	Stores internship applications, including company name, position, deadline, status, and notes.
documents	Stores uploaded CV, transcript, and certificate links from Amazon S3.
ai_analysis_results	Stores AI analysis output such as CV text, job description, matching score, matched skills, missing skills, feedback, and creation time.
interview_questions	Stores AI-generated interview questions for each application or job description.

6.1 Example Fields for `ai_analysis_results`

- `id`
- `application_id`
- `cv_text`
- `job_description`
- `match_score`
- `matched_skills`

- `missing_skills`
- `ai_feedback`
- `created_at`

7 Eight-Week Development Timeline

Week	Work
1	Learn AWS web cloud architecture.
2	Build the basic internship tracker web application.
3	Deploy backend/frontend to AWS and connect Amazon S3.
4	Connect Amazon RDS database.
5	Add CloudWatch logging and error monitoring.
6	Add AI CV-Job Matching feature.
7	Test matching score, improve UI, and optimize performance.
8	Final documentation, product demo, evaluation, and summary.

8 Expected Outcome

At the end of the project, the team should have a working cloud-based web application deployed on AWS. The system should allow users to manage internship applications, upload documents, analyze CV-job fit using AI, and monitor system behavior through CloudWatch. This makes the project suitable for demonstrating both cloud application development and practical AI integration.